

Operating Systems and Kernel Design

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EECE 4811/EECE 5811

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Assignment #2

This second assignment's focus is on threads and their synchronization. You will make use of pthreads on Linux to implement your assignment.

1. You are asked to implement a simple main program with 2 (two) threads that we call "*providers*" and 260 (two-hundreds and sixty) threads that we call "*buyers*". The *providers*' threads produce different integer numbers and put them in a buffer. Design and implement a program that will handle the synchronization of the *providers*' threads as well as the *buyers*' threads. Basically, the program should make sure that the *providers* repeatedly insert items (here integers identifying/representing items) and prints them on the screen, and also makes sure only one *buyer* at a time buys an item (represented by an integer) and prints it on the screen. The other *buyers* (threads) should sleep for some time while the thread *buyer* is buying. When a thread wakes up, it tries to *buy* an integer from the buffer. Make sure that each and all *buyers* get a chance to buy an item represented by a number. No *buyer* should make the other ones starve. The *providers* should not stop providing items (i.e., numbers representing items). Make sure that the *providers* get a chance to insert items (i.e. numbers) in the empty buffer locations where the buyers buy items from non-empty buffer locations. Make sure that each buyer can buy an item from a non-empty buffer location. Only one thread from either the provider thread or the 260 buyer threads can access a buffer location at a time.

Your program should have one command-line argument which specifies the number N of *buyer* threads. When the program starts, it creates $N+2$ threads which place themselves in a thread pool. The main program issues the initial notification to the *provider* threads to start providing integers and then notifies the main program to start doing its job as described above by making sure that the N *buyer* threads have a chance to buy items (represented by numbers).

2. After keeping a copy of program-1, from part-1 above (of course, that will be needed in the report for part-1), modify the program of part-1 and run it in the case of *one single provider* and *6 buyers*. Explain in your report what will be one main significant change to your code to make it efficient, if any.